

# Chris Rockwell

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| PROFESSIONAL EXPERIENCE | <b>Roblox</b><br><i>AI Scientist, Foundational AI</i><br>• Multimodal generative AI towards photorealistic gaming<br><b>Morpheus AI</b><br><i>Member of Technical Staff</i><br>• Led data curation and evaluation for real-time video and world models<br><b>NVIDIA</b><br><i>Research Intern, Deep Imagination Research Group</i><br>• Systematically curated and annotated cameras on 100K diverse, dynamic Internet videos<br><b>Meta</b><br><i>Research Scientist Intern, Computational Photography Research</i><br>• Produced lightweight radiance field conditioned upon a single image<br><b>TuringSense</b><br><i>Technical Consultant, Computer Vision</i><br>• Suggested and implemented improvements to TuringSense home yoga product<br><b>Citadel</b><br><i>Trader, Global Fixed Income</i><br>• Designed and executed quantitative trading strategies to enhance team's portfolio<br><b>BNP Paribas</b><br><i>Analyst, Fixed Income Structuring</i><br>• Created systematic hedging strategies and priced bespoke options for institutional clients | San Mateo, CA<br>Jun. 2026 - Present<br><br>Pittsburgh, PA<br>Sep. 2025 - May 2026<br><br>Santa Clara, CA<br>Mar. 2024 - Oct. 2024<br><br>Seattle, WA<br>May 2022 - Dec. 2022<br><br>Santa Clara, CA<br>Feb. 2021 - Apr. 2021<br><br>New York, NY<br>Apr. 2017 - Oct. 2017<br><br>New York, NY<br>Jul. 2015 - Mar. 2017 |
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| EDUCATION | <b>University of Michigan</b><br><i>Ph.D. in Computer Science and Engineering</i><br>• Advisors: Justin Johnson, David F. Fouhey<br><br><i>Master of Science, Computer Science and Engineering</i><br>• GPA: 4.00/4.00<br>• Advisors: David F. Fouhey, Jia Deng<br><br><i>Bachelor of Science, Economics</i><br><i>Minors in Computer Science, Mathematics</i><br>• GPA: 3.95/4.00 | Ann Arbor, MI<br>Sep. 2020 - Aug. 2025<br><br>Sep. 2018 - May 2020<br><br>Sep. 2011 - May 2015 |
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| INTERESTS | Computer Vision, Machine Learning |
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| PUBLICATIONS | Dynamic Camera Poses and Where to Find Them<br><b>Chris Rockwell</b> , Joseph Tung, Tsung-Yi Lin, Ming-Yu Liu, David F. Fouhey and Chen-Hsuan Lin<br>CVPR, 2025<br><a href="#">Project Page</a> |
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FAR: Flexible, Accurate and Robust 6DoF Relative Camera Pose Estimation  
**Chris Rockwell**, Nilesh Kulkarni, Linyi Jin, JJ Park, Justin Johnson and David F. Fouhey  
CVPR, 2024 (**Highlight**)

[Project Page](#)

Scalable 3D Captioning with Pretrained Models  
Tianghe Luo\*, **Chris Rockwell**\*, Honglak Lee<sup>†</sup> and Justin Johnson<sup>†</sup>  
NeurIPS (Datasets and Benchmarks Track) 2023

[Project Page](#)

The 8-Point Algorithm as an Inductive Bias for Relative Pose Prediction by ViTs  
**Chris Rockwell**, Justin Johnson and David F. Fouhey  
3DV 2022

[Project Page](#)

PlaneFormers: From Sparse View Planes to 3D Reconstruction  
Samir Agarwala, Linyi Jin, **Chris Rockwell** and David F. Fouhey  
ECCV 2022

[Project Page](#)

FWD: Real-time Novel View Synthesis with Forward Warping and Depth  
Ang Cao, **Chris Rockwell** and Justin Johnson  
CVPR 2022

[Project Page](#)

Understanding 3D Object Articulation in Internet Videos  
Shengyi Qian, Linyi Jin, **Chris Rockwell**, Siyi Chen and David F. Fouhey  
CVPR 2022

[Project Page](#)

PixelSynth: Generating a 3D-Consistent Experience from a Single Image  
**Chris Rockwell**, David F. Fouhey and Justin Johnson  
ICCV 2021

[Project Page](#)

Full-Body Awareness from Partial Observations  
**Chris Rockwell** and David F. Fouhey  
ECCV 2020

[Project Page](#)

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RESEARCH  
EXPERIENCE

**Morpheus AI**

Pittsburgh, PA

*Member of Technical Staff* | Founder: Xun Huang

Sep. 2025 - May 2026

Real-Time Video Generation

- Led data curation and evaluation for real-time video and world models

**NVIDIA, Deep Imagination Research Group**

Santa Clara, CA

*Research Intern* | Hosts: Chen-Hsuan Lin, Tsung-Yi Lin

Mar. 2024 - Oct. 2024

Internet Scale 3D Curation – *Dynamic Camera Poses*

- Curate and annotate cameras for 100K dynamic Internet videos from 3.2M diverse videos

**Meta Reality Labs, Computational Photography Research**

Seattle, WA

*Research Scientist Intern* | Hosts: Hung-Yu Tseng, Jia-Bin Huang

May 2022 - Dec. 2022

Novel View Synthesis

- Produce lightweight radiance field conditioned upon a single image

**Michigan Vision Lab***Graduate Research Assistant* | Advisor: Justin JohnsonAnn Arbor, MI  
May 2020 - Aug. 2025Internet Scale 3D-Text Modeling – *Scalable 3D Captioning*

- Apply powerful VLM pipeline to caption 660K 3D assets, finetune text-to-3D models

Novel View Synthesis

- *PixelSynth*: Introduce powerful generative model, enabling 3D-consistent extrapolation
- *FWD*: Real-time NVS using pointcloud and transformer; predecessor to 3DGS

**Fouhey AI Lab***Graduate Research Assistant* | Advisor: David F. FouheyAnn Arbor, MI  
May 2019 - Aug. 2025

Relative Camera Pose Estimation

- *8-Point ViT*: Include 8-Point machinery in ViT block to improve relative pose estimation
- *FAR*: Fuse correspondence and learning-based pipeline, yielding best-of-both estimates

3D Reconstruction – *PlaneFormers*

- Introduce transformer to learn to refine planar reconstruction

3D Object Articulation – *Understanding 3D Object Articulation*

- Collect rich dataset of people articulating objects and learn axes of object articulation

3D Human Pose Estimation – *Full-Body Awareness*

- Propose self-training method to substantially improve human pose on internet video

**Princeton Vision and Learning Lab***Graduate Research Assistant* | Advisor: Jia DengPrinceton, NJ  
May 2018 - May 2019

2D Human Pose Estimation

- Add bottleneck-to-attention module to improve *Stacked Hourglass* accuracy 0.7%

Meta-Learning

- Improve finetune model to within 0.1 *avg. rank* of meta-learning baseline on *Meta-Dataset*

**Strategic Reasoning Group***Undergraduate Research Assistant* | Advisor: Michael P. WellmanAnn Arbor, MI  
May 2013 - Jul. 2013

Agent-based simulation of High-Frequency Trading and Latency Arbitrage

- Model trading agents with varying speeds to measure effects of latency arbitrage

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**SERVICE**

Reviewer: CVPR (2023 Outstanding Reviewer), NeurIPS (2023 Top Reviewer), ICCV, ECCV, 3DV, ICLR, ICML, TPAMI

AI4ALL Project Instructor: lead vision project for nine underrepresented high-schoolers

AI4ALL Curriculum Advisory Board Member: contributed to national curriculum

Technical Mentor: mentored five students with David F. Fouhey, including one in African Undergraduate Research Adventure (AURA)

Graduate Student Advisory Committee: represented CSE students to improve experience